PAPER_1202_UQFF_Quantum_Chain_E_n_Summation_633333_Validation

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PAPER_1202 — UQFF Quantum Chain E_n Summation & 633333.333 Validation

Sole Canonical Source (dpm_vacuum_manifold.py:12) ``` THE QUANTUM CHAIN (canonical, IMMUTABLE): Step 0 0_vacuum -> |grad(UA)| vacuum tension differential Step 1 grad(UA) -> DPM_vortex a_DPM = F_DPM*f_DPM*E_vac/(c*V_sys) Step 2 DPM_vortex -> mu_s mu_s = rho_A * V_DPM Step 3 mu_s -> Ug1[seed=DPM] Ug1 seeded from mu_s -- NOT from mass Step 4 Ug1 -> Ug_family Ug2+Ug3+Ug4 simultaneously promoted Step 5 Ug_family -> F_U + Um + FUBi + FUBii + UA_uv Step 6 F_U -> crossing FUBi(r) + FUBii(r) = 0 compaction Step 7 crossing -> M_emergent mass BORN at crossing, not before Step 8 M_emergent -> GM/r^2 LAST -- observational projection only ```

Compatibility Note `dpm_vacuum_manifold.py` remains the canonical consolidated source file. The restored legacy modules `scm_vacuum_manifold.py` and `ua_vacuum_manifold.py` are available again for compatibility, explicit SCm/UA layer inspection, and delegated numeric helper calls when importable.

derive_from_quantum_chain (dpm_vacuum_manifold.py:74) ```python def derive_from_quantum_chain(n_levels=26, f_SCm=0.57, V=1.0): E_n = [E0 * 10n for n in range(1, n_levels + 1)] rho_vac_energy = sum(f_SCm * E for E in E_n) / V # J/m³ rho_mass_eq = rho_vac_energy / (_C_LIGHT ** 2) # kg/m³ (gravity only) return rho_vac_energy, rho_mass_eq ```**

Structural Constants (dpm_vacuum_manifold.py:97) ``` E0 = 1e-20 SSQ = 0.57 KAPPA = 5e-4 THZ_PHONON = 1.25e12 RHO_VAC_SCM = 4.0 * math.sqrt(math.pi) * 1.0e-37 # 7.0898154036e-37 J/m³ RHO_VAC_UA = 10.0 * RHO_VAC_SCM ```

E_n Summation + F_U_Bi_i_99 (dpm_vacuum_manifold.py:80,132) ``` E_n = [E0 * 10n for n in 1..26] rho_vac_energy = sum(0.57 * E for E in E_n) / V F_U_Bi_i_99 = Sum(-BETA_I * Ug_k * cos(pi*t_n) * (M / r**2), (k,1,99)) ```**

Live Validation (2026-05-27 execution) ``` QuantumChain direct (derive_from_quantum_chain(26,0.57)): rho_vac_energy = 633333.3333333334 J/m^3 DERIVATIONS.derive_condensed_effective_rho_scm(): 633333.333 (exact target match, abs diff < 1e-6) RHO_VAC_SCM_micro = 7.0898154036e-37 RHO_condensed (S26_3 amp) = 633333.333 ```

10 derive_* Consume Quantum Chain Solely (_uqff_primitives.py:746,681) ``` derive_condensed...: rho_cond = RHO_micro * S26_3 * PHI_RES * (1+KAPPA*1e4) * geom derive_G_newton: uses rho_scm (Quantum Chain micro) + S26_3 (26D from E_n) derive_beta_i / hbar / masses / alpha / c / V_SCM / hz: all chain through rho or S26_3 All 10: zero external seeds, sole source dpm v3.0 Quantum Chain + 26D origami ```

Cross-References - dpm_vacuum_manifold.py:74 (E_n sole source) - _uqff_primitives.py:817 (DERIVATIONS uses rho from chain) - QCalcGeom.py:18 (QUANTUM CHAIN compliance) - MAIN_1:2852 (F_U from Ug_family + FUBi/FUBii at crossing) - LEDGER_FIRST_PRINCIPLES_DERIVATIONS.md:94 (Quantum Chain mathematical equation) - WhitepaperGapAnalysis.md:101 (G4 gap closure)